Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Valentin Richter Trummer	**Email**	Valentin.Richter-Trummer@iwu.fraunhofer.de
Institute Name	Fraunhofer IWU			
Address	Reichenhainer Str. 88, 09126 Chemnitz, Germany			

2. Technology Overview

Title of Proposed Technology	**AIRMAN** (**a**ikido **i**nspired **r**obotic **man**ipulators) - novel methods for autonomous robotic manipulation -			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (.), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical Materials		*Mechanical Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics / Automation Machinery	*Robot Vision and Production Automation Technologies*	*Robotics / Automation Machinery*	*Robot Control and Intelligence Technologies*

Concept of Proposed Technology	*Automated disassembly is currently too expensive for broad application. Novel solutions for maximizing the applicability of adaptive endeffectors can optimize this situation. The proposed endeffectors harmonically and autonomously adapt to the workpiece. For this purpose a set of solutions is proposed in the following two main areas, which make use of todays sensor and control capabilities:*
	• *Logistics:*
	  ○ *Force driven disassembly (e.g. for battery covers and battery modules) – autonomous path of the weakest resistance*
	  ○ *Gripper system with autonomous local adjustment (local searching for screws/holes/…) empowered by AI-supported image processing*
	• *Process:*			

	○ *Endeffectors for autonomous cleaning/sanding/grinding (e.g. for paint or adhesive removal)* ○ *Multiphysics-based (e.g. heat and/or vibration) adhesive breaking integrated into disassembly endeffector system* *The combination of all or part of the above proposed solutions shall allow creating flexible an affordable robotics solutions, which due to their adaptability can be used for a significant amount of different applications, therefore justifying investing into automation even in challenging environments like disassembly.*
Rationale for Joint R&D Project	*Korean KMUs have significant Knowhow and experience with disassembly of components in the electromotive market. Furthermore very advanced capabilities exist in the area of sensing and edge processing. The Fraunhofer IWU has significant experience in efficiently automating complex processes. Together it can be possible to develop automated solutions compatible with the market requirements.*
Key Words	1. *autonomous process* 2. robotic endeffectors 3. process force 4. Economic and time assessment using simulation tools 5. AI-supported image processing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	• *Advanced multi-physics based control strategies for industrial robots* • *Adaptable robotics endeffectors* • *Development of robust and inexpensive sensorial solutions based on sensor data fusion*			
R&D Target Specifications	• The goal is to develop a laboratory demonstration of a multi-physics based robotics control strategy for complex disassembly scenarios, including the required surrounding processes. By the end of the project it is expected that the feasibility of the proposed approach is demonstrated and an industrial solution can be specified			
R&D Plan				

WP1	Requirement definition	Milestone 1: Conditions, expectations, and boundaries are defined.
WP2	Concept development	Milestone 2: The demonstrator concept was developed jointly with a focus on simulating cycle times and estimating costs prior to implementation.
WP3	Laboratory level demonstration	Milestone 3: The sample components provided by the industry partners were disassembled in the developed demonstrator. This refers to the successful implementation of the previously defined disassembly processes (e.g., unscrewing process, cutting process, pressing or pulling processes)
WP4	Specification of industrial solution	Milestone 4: The necessary steps for industrial qualification for commercialization are defined and evaluated in terms of the necessary time and financial expenditure. An estimation of the actual costs required to implement an automated dismantling solution and the necessary number of components to be dismantled is calculated in order to guarantee economic feasibility.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• AI based image processing (especially focusing on extracting the best possible geometrical information from the smallest and lowest cost sensors possible) • Development of low-cost but high quality load cells for force driven process execution, including robust signal processing for maximizing the SNR at the lowest possible cost and size
Expected R&R of Korean SME Partner	• Use-case provider, especially also in the initial requirements definition phase as well as the final validation. • Test material shall be provided for disassembly trials • Sharing expertise of the currently both manual and semi-automated process execution in the company → sharing demands for automated implementation

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Laboratory demonstrator of a multi-physics-based robotics control strategy for complex disassembly scenarios, including the required surrounding processes. The commercialization potential lies in political regulations like the "EU-Altfahrzeugverordnung" - a measure that obliges OEMs to increase the proportion of recycled material in new production. Global, social, and political tensions such as wars, trade conflicts, and the artificial scarcity of raw materials for strategic reasons lead to logistical and financial planning uncertainty. Diversifying sources of primary and secondary raw materials promotes corporate resilience and thus competitiveness. The envisaged solution therefore paves the way for economical feasible automated and therefore flexibly scalable disassembly.

6. References

The proposed technology modules are meant for disassembly tasks but are also partly qualified for assembly tasks by the Fraunhofer IWU:

a) *https://www.iwu.fraunhofer.de/de/projekte/zirkel-zirkulaere-produktion-fuer-hochintegrierte-komponenten-der-elektromobilitaet.html; 14.01.2026*

b) *https://www.iwu.fraunhofer.de/de/projekte/die-zukunft-des-autorecyclings-wiederverwendung-statt-verschrottung.html; 14.01.2026*

c) *https://www.iwu.fraunhofer.de/de/projekte/tavieda-integrierte-produkt--und-prozessentwicklung-fuer-die-naechste-generation-von-flugzeugtueren.html; 14.01.2026*

d) *https://www.iwu.fraunhofer.de/de/projekte/lernende-roboterschleiftechnik-fuer-das-handwerk-lerosh.html; 14.01.2026*

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization